



A64FX Architecture: a case study

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Three Pillars of Performance



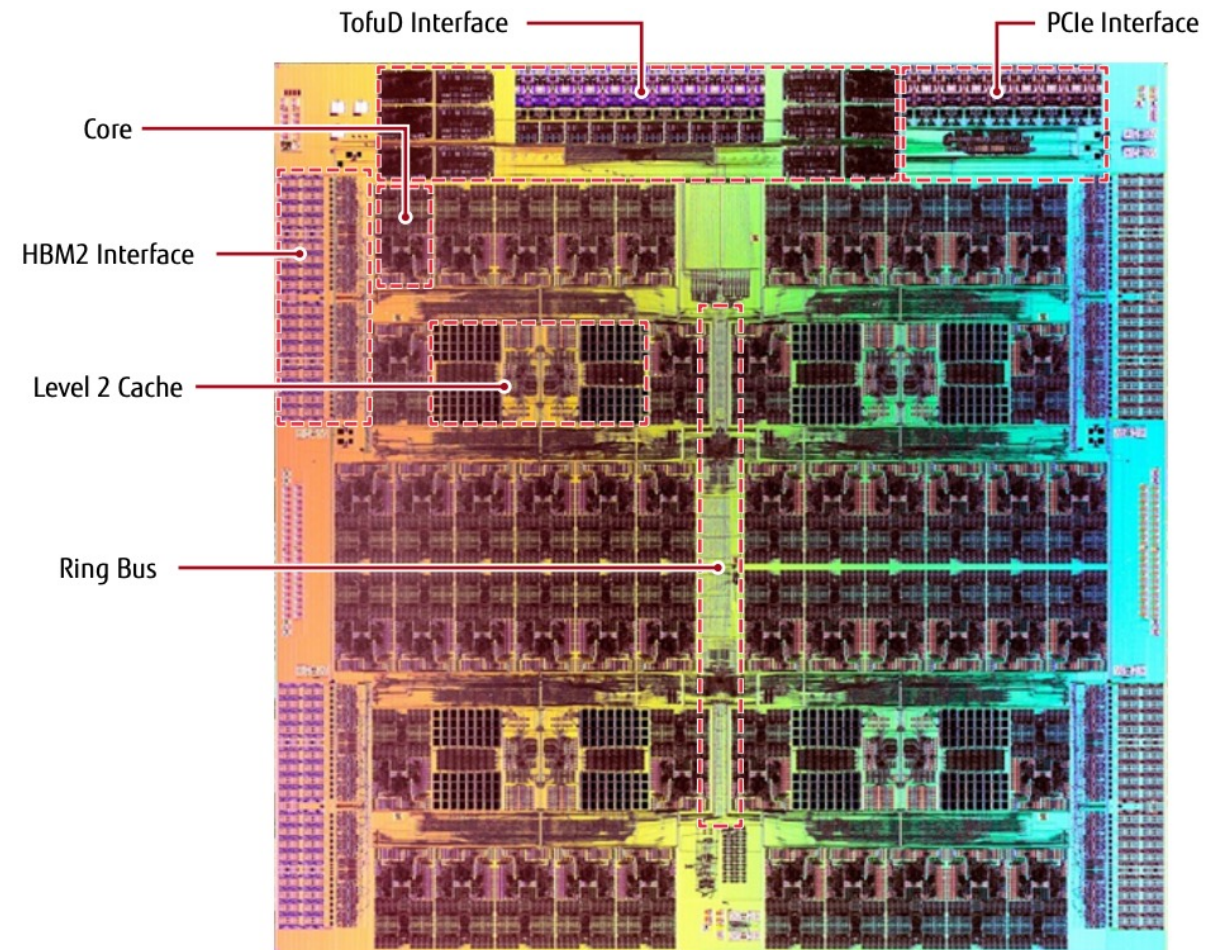
- **High-Bandwidth Memory**
 - 32 GiB of High Bandwidth Memory (HBM2) on-package, delivering 1 TiB/s of memory bandwidth
- **High-Performance Vectorization**
 - SVE ISA with a massive 512-bit vector width, tailored for scientific and AI kernels
- **Massive Parallelism**
 - Many-core design, integrating 48 high-performance compute cores



The A64FX Die



ISA	Armv8.2-A + Scalable Vector Extension (SVE)
Process Technology	7nm FinFET
Transistor Count	~8.8 billion
Core Count	52 total: 48 Compute Cores + 4 Assistant Cores
Chip Topology	4 Core Memory Groups (CMGs); 4-node ccNUMA
Frequency	1.8 GHz / 2.0 GHz / 2.2 GHz
Peak DP FLOPS	3.3792 TFLOPS (at 2.2 GHz)
Peak SP FLOPS	6.7584 TFLOPS (at 2.2 GHz)
Peak HP FLOPS	13.5168 TFLOPS (at 2.2 GHz)
Main Memory Type	HBM2 (3D Stacked, On-Package)
Memory Capacity	32 GiB
Memory Bandwidth	1,024 GB/s
L2 Cache	32 MiB (4 x 8 MiB)
Interconnect	Tofu Interconnect D (Integrated)
I/O	PCIe Gen3 x16
Derived BF Ratio (DP)	~3.2 FLOPS/Byte

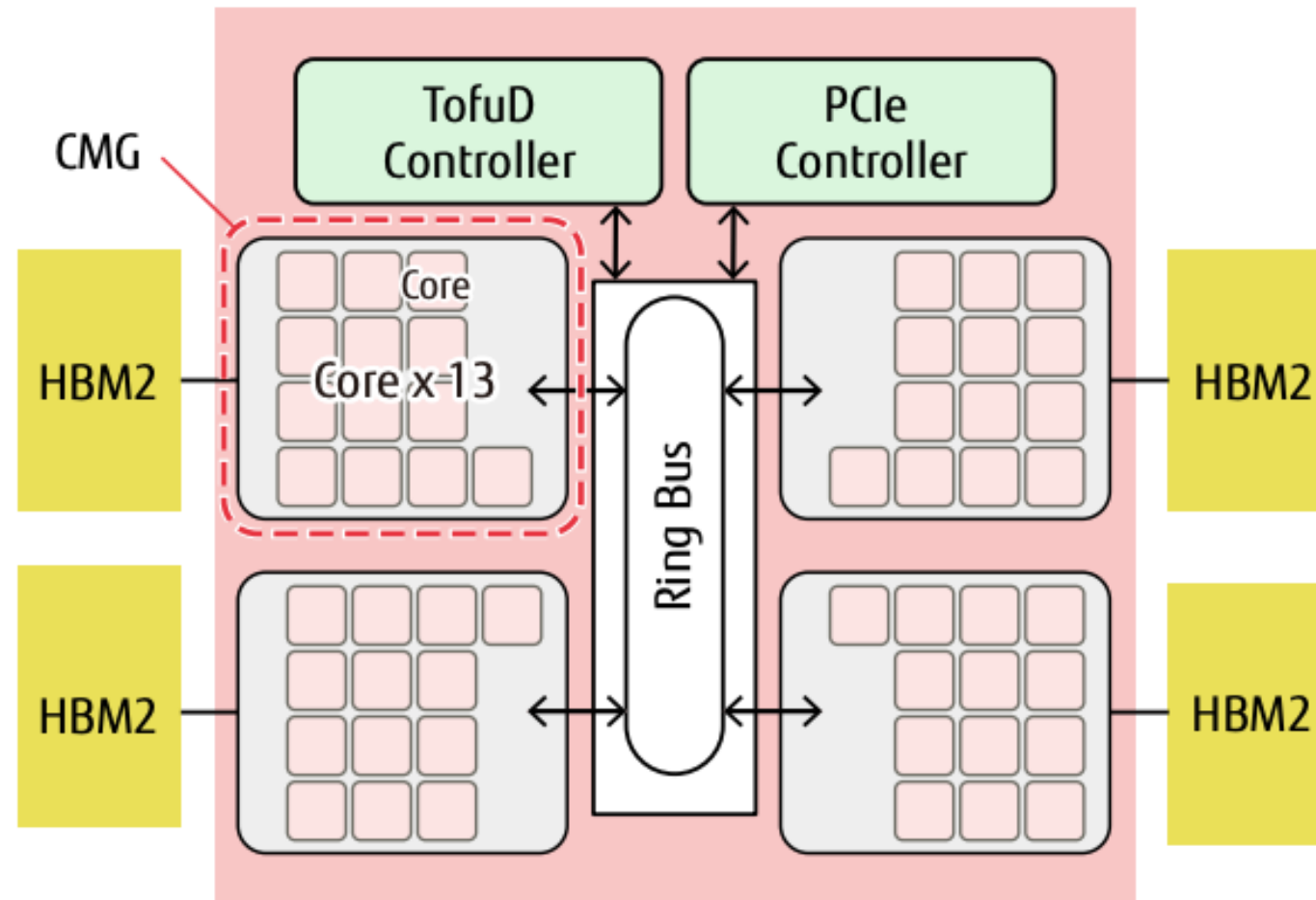


A Block Diagram of the Die



- A single CMG consists of:

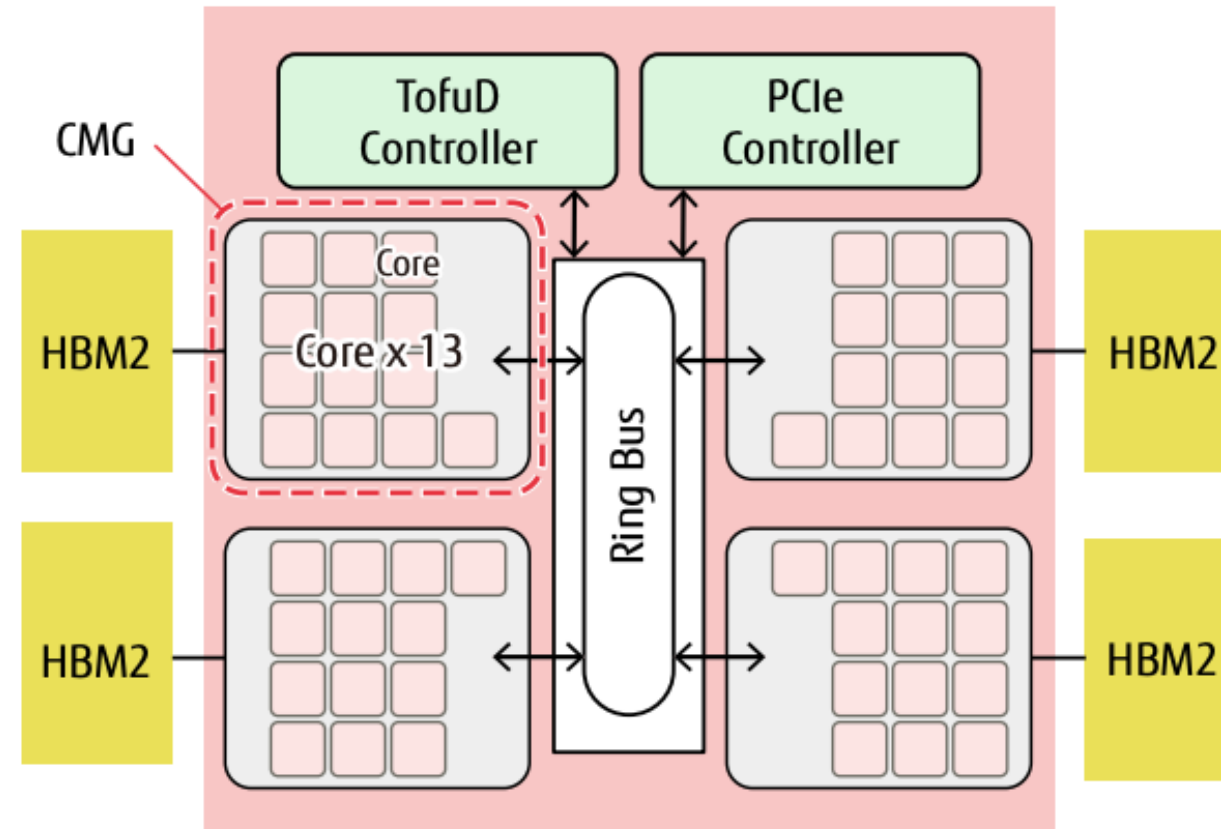
- 13 identical processor cores
- 8 MiB, 16-way set-associative unified L2 cache. This cache is shared by all 13 cores within the CMG.
- HBM2 Memory Access Controller (MAC). This controller has a direct connection to one of the four HBM2 stacks.
- The 13 cores within the CMG are connected to the shared L2 cache via a high-throughput X-bar connection.



A Block Diagram of the Die



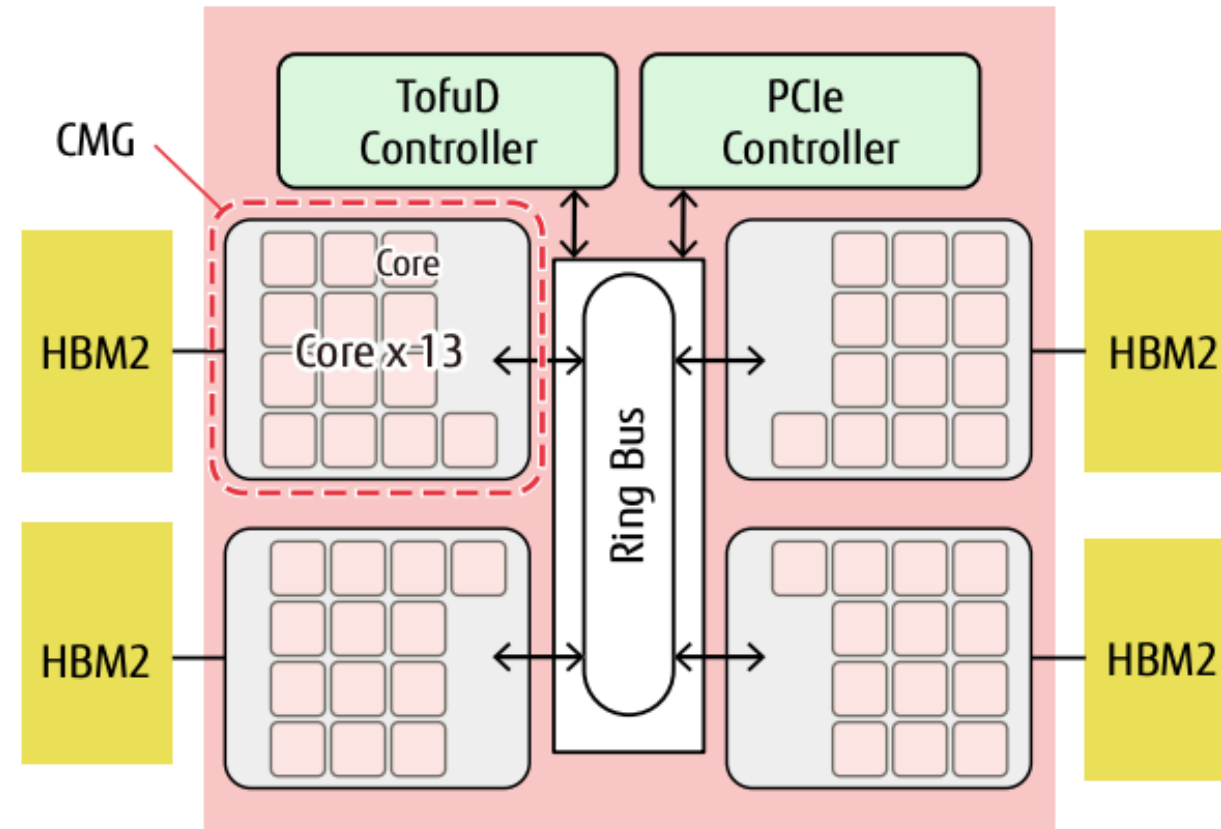
- The assistant cores, identical to the compute cores, are **reserved for "non-computational purposes"**
 - hardware-level mechanism for mitigating "OS jitter"
 - In massively parallel systems, application performance can be degraded by system noise
 - Mitigates thread interruption by the OS, especially to handle the TofuD interconnect



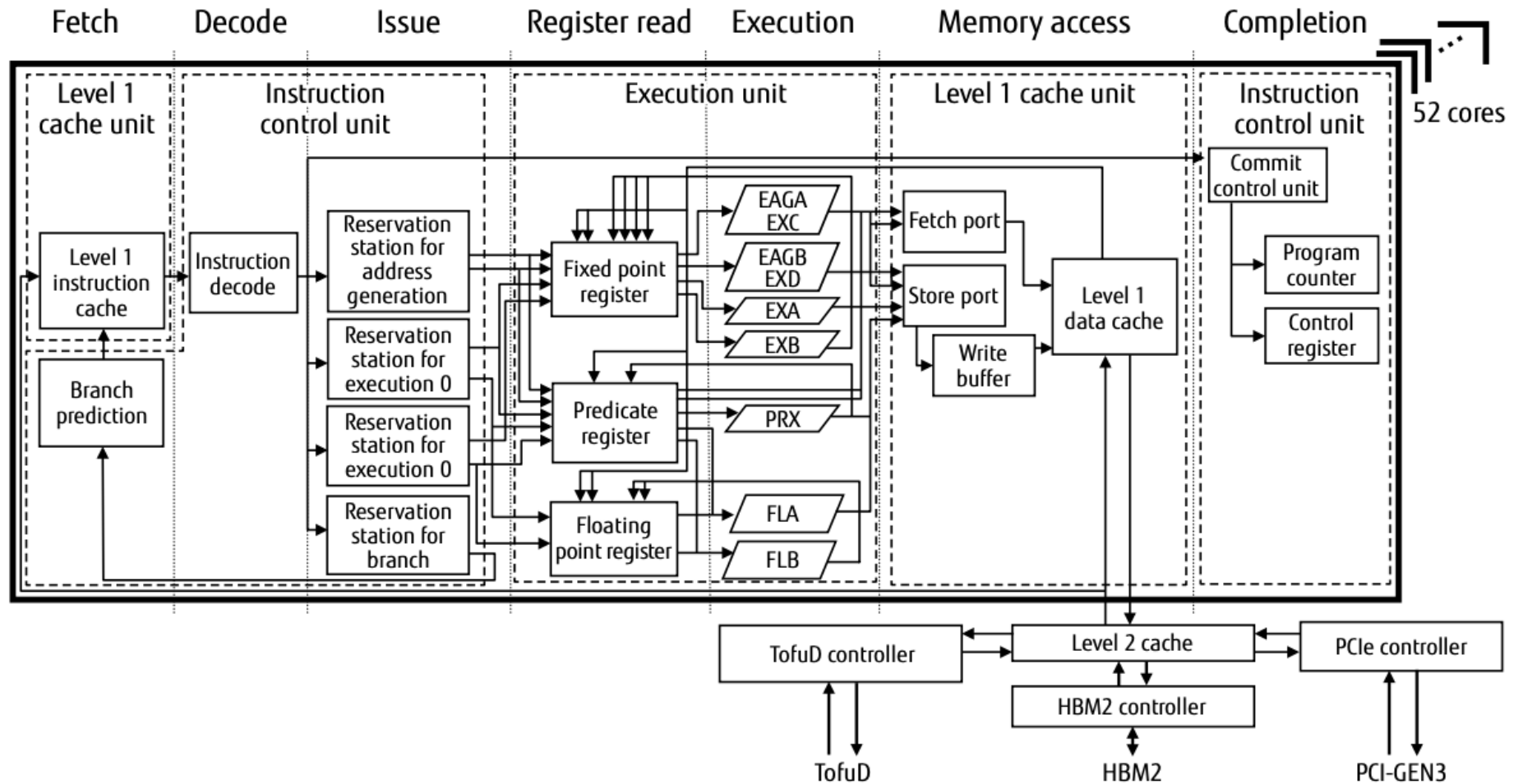
On-Chip Interconnection



- The Network on Chip (NoC) is based on a hybrid ring-mesh interconnection
 - Main ring bus + proprietary "CMG interconnect paths" that provide direct, high-speed links between *adjacent* CMGs.
 - Ring bus provides full connectivity among CMGs.
 - Direct adjacent-CMG paths (throughput over 115 GiB/s) provide a low-latency, high-bandwidth shortcut for NUMA-local traffic.



Instruction Pipeline Overview

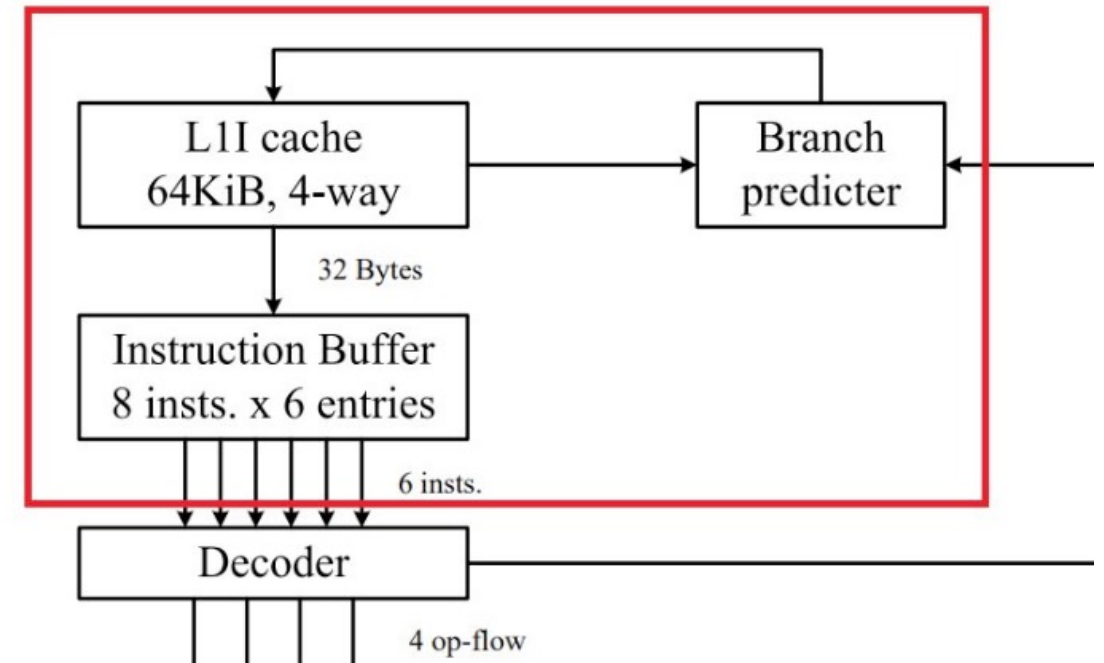




Fetches 8 instructions simultaneously

Predicts four branch directions per cycle

Predicts direction of one taken branch per cycle



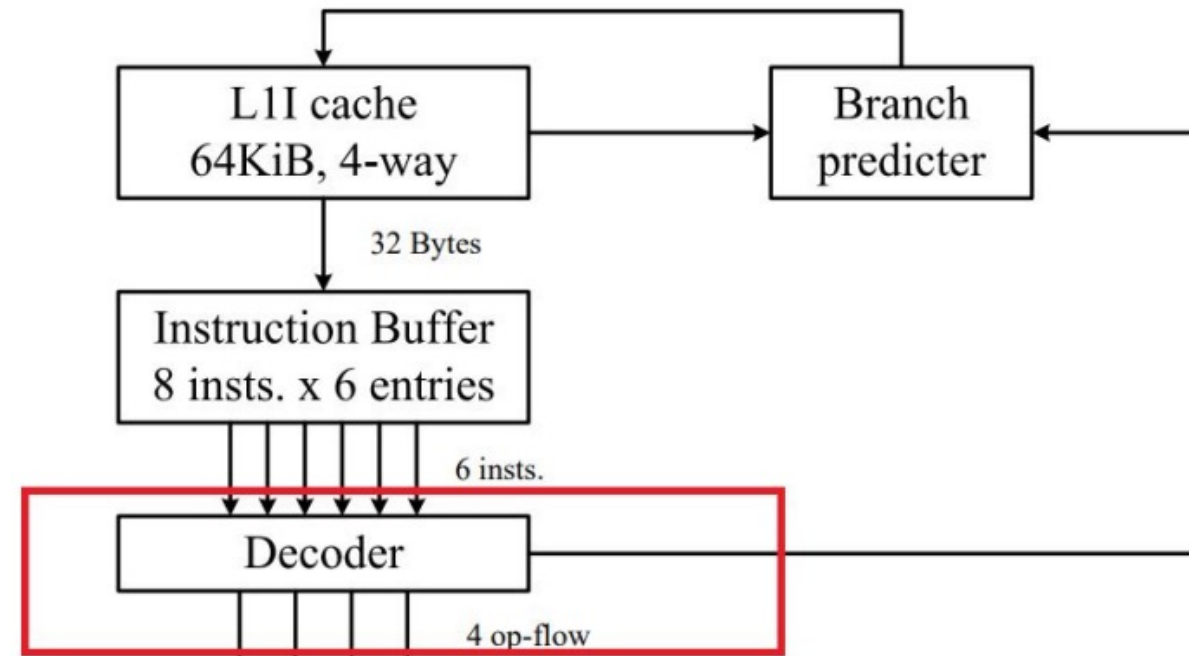
Source: [A64FX Microarchitecture Manual v1.8](#)



Takes in up to 6 ARMv8 instructions per cycle

ARMv8 instructions decoded into μ OP instructions

Broken down μ OPs sent in-order to execution stage



Source: [A64FX Microarchitecture Manual v1.8](#)

Execution - Reservation Stations



Int/floating/predicate

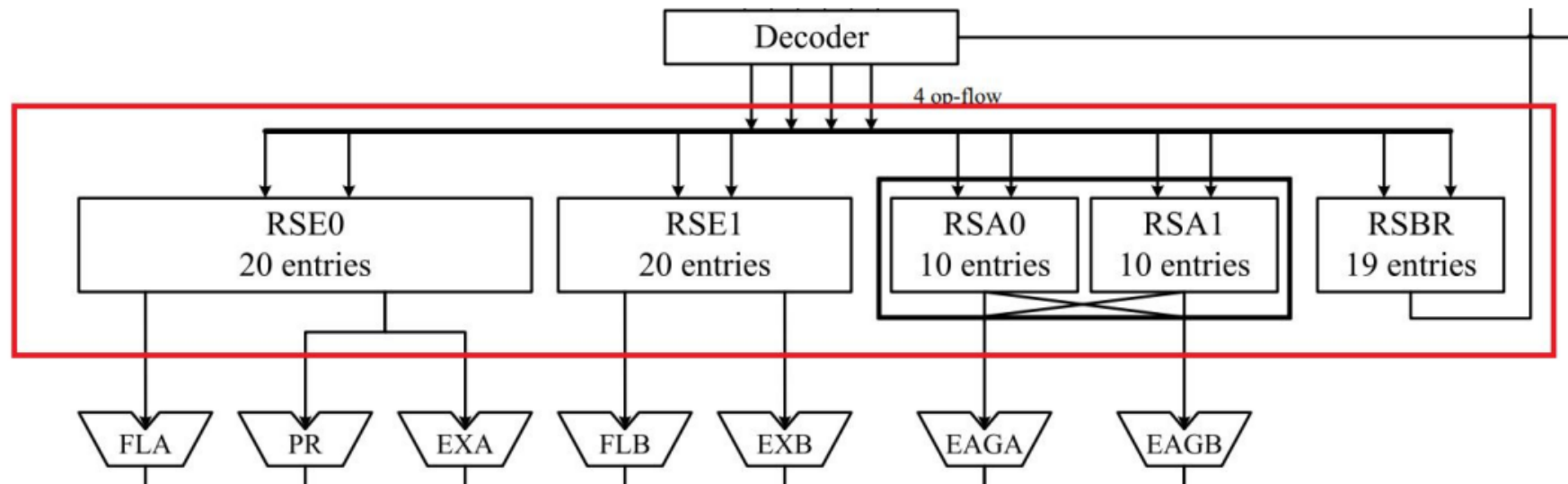
Address calculation

Branch prediction

2 stations
20 entries each

2 stations
10 entries each

1 station
19 entries

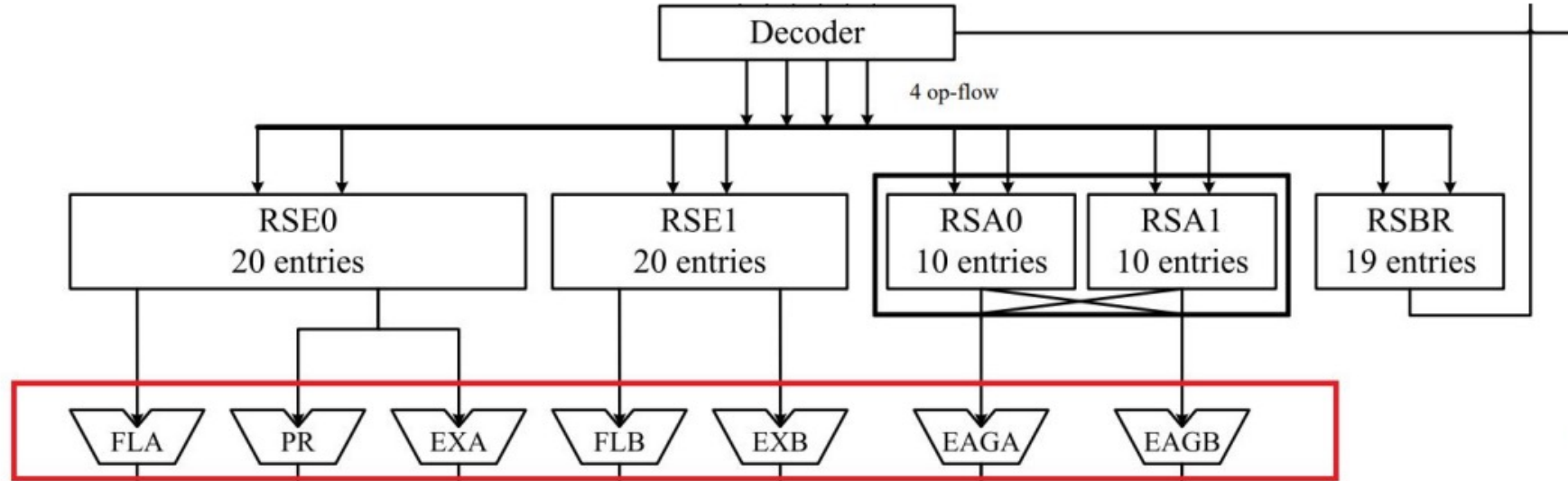


Source: [A64FX
Microarchitecture
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Five separate pipelines for integer, SIMD floating point and SVE, predicate, branch, and load/store operations respectively

Pipelines are 11-24 stages deep and support operand bypassing



Source: [A64FX Microarchitecture Manual v1.8](#)

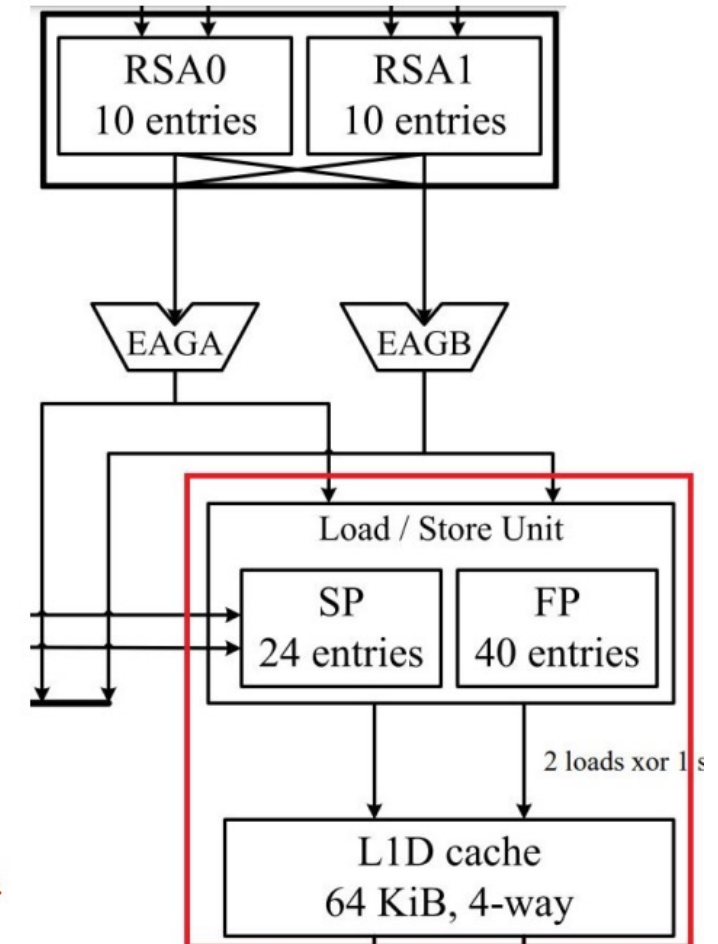
Load and Store Data



Executes one store or two loads at once

Uses virtual load and store ports

Load/store both pipelined for performance



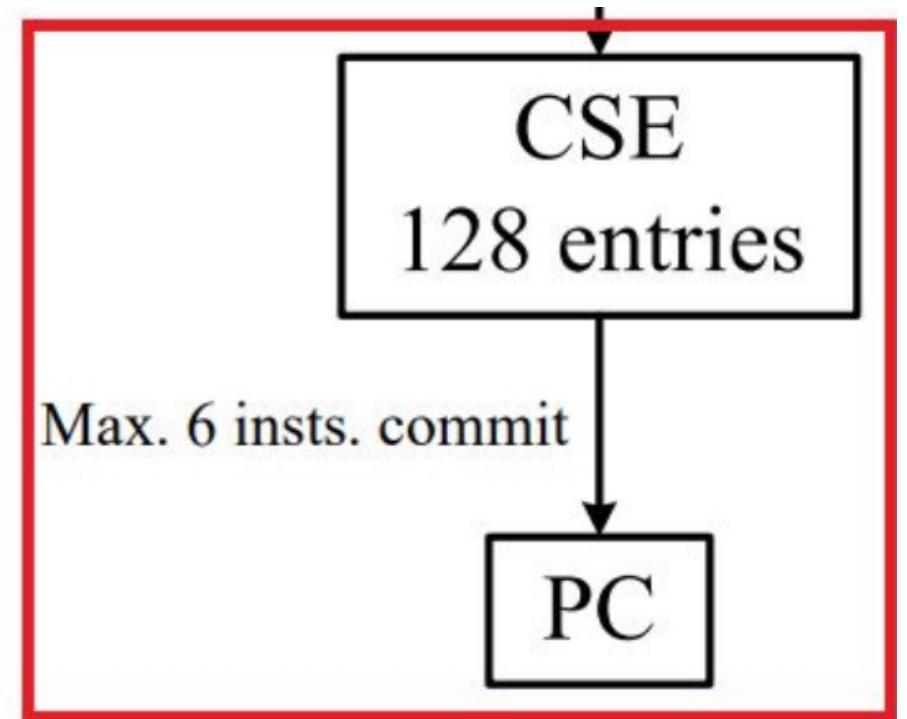
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Uses a commit stack to maintain proper execution

Checks exceptions and branch prediction results

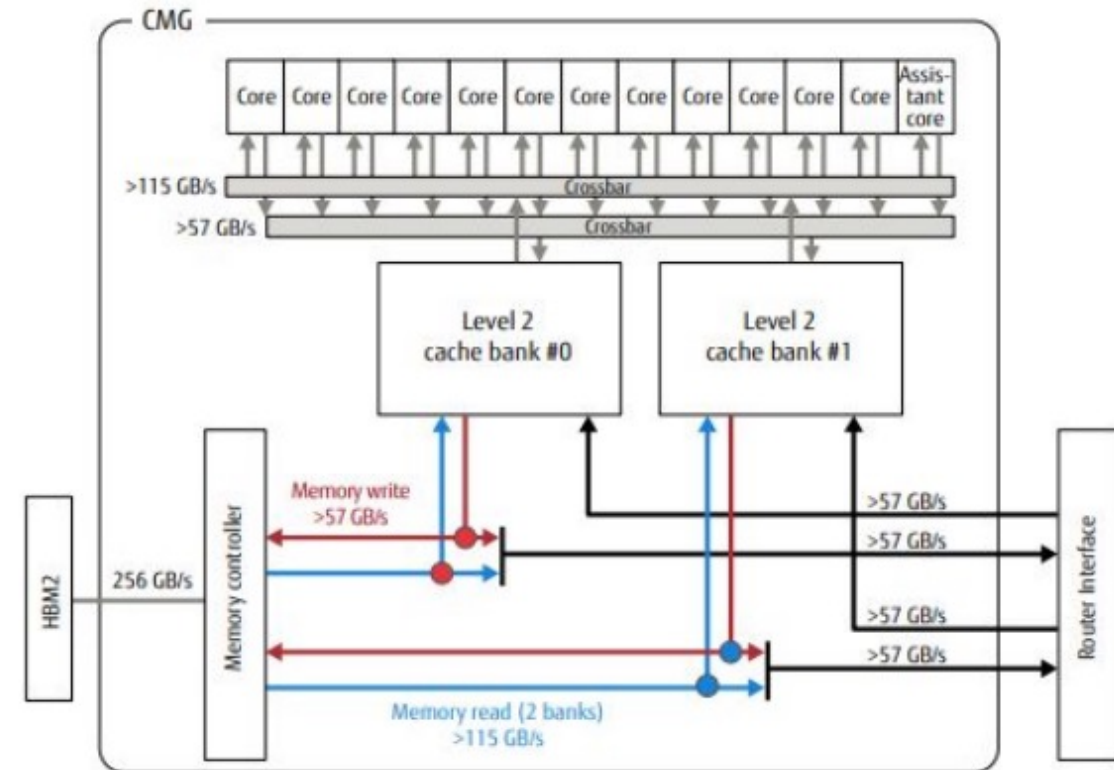
Commits 4 μ OPs per cycle



Memory Hierarchy



- The processor employs Single Instruction Multiple Data Compute method with 512 bits as the width.
- L1 I-Cache is 3MiB (64KiB/Core)
- L1 D-Cache is 3MiB (64KiB/Core)
- L2 Cache is 32MiB (8MiB*4)
- This 4x multiplier is due to the CMGs that this processor has (*CMG is Core Memory Groups).
- The Memory Accesses between these CMGs is in NUMA config.





- **Manufacturing**
 - 7nm FinFET process - low power consumption and high transistor density.
- **Microarchitecture**
 - Dynamic clock gating of execution units.
 - Short Loop Detector that powers down the entire front-end during tight compute loops.
 - Simple memory hierarchy - no DRAM controllers; no power-hungry L3 cache.
- **Power Knobs**
 - A software-accessible function that allows an application to dynamically control hardware.
 - Software to *disable* specific execution pipelines.
 - A memory-bandwidth-bound application can have its runtime (or the user) programmatically power-gate its unused floating-point pipelines or compute cores. **Lots of research on doing this dynamically.**



- The A64FX is the first processor to commercially implement the Arm Scalable Vector Extension (SVE).
 - SVE is a SIMD instruction set, co-designed with Fujitsu, intended to replace the 128-bit NEON ISA for HPC and AI workloads.
 - A **key distinction** of SVE is its "Vector-Length Agnostic" (VLA) design.
 - Binaries compiled with SVE can run, **without recompilation**, on any SVE-compliant hardware.
- The A64FX supports vector lengths of 128, 256, and 512 bits.
 - Physical data path and registers are **512 bits wide**.
- A large **256-byte** line size for both L1 and L2 caches.

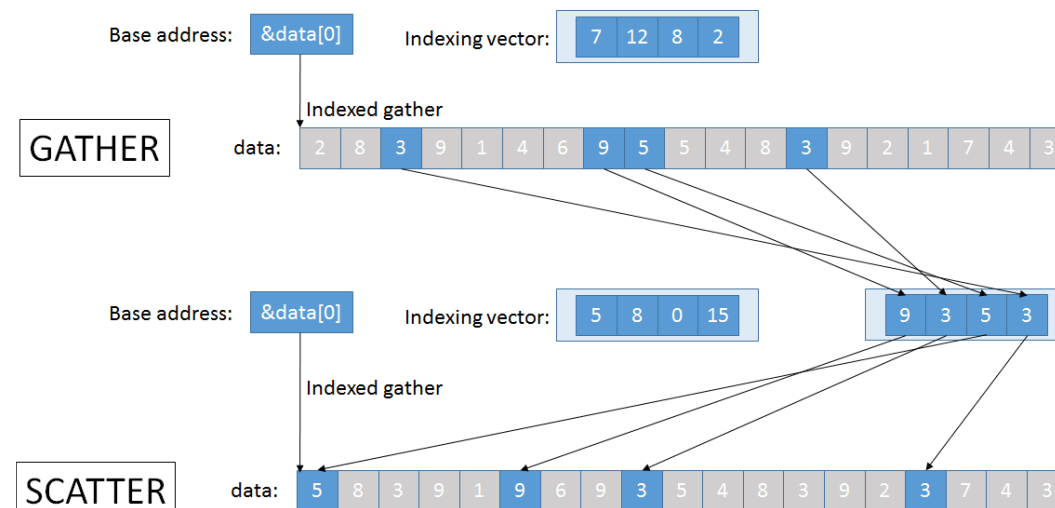


- **Vector Registers**
 - SVE provides 32 vector registers (Z0-Z31), which are 512 bits wide in the A64FX.
- **Predicate Registers (PPRs)**
 - It provides 16 predicate registers (P0-P15). On A64FX, these are 64 bits wide (one bit per byte of the 512-bit vector).
- **Predicated Execution:**
 - Most SVE instructions are *predicated*, where every operation is governed by a predicate register, which acts as a "mask".
 - Only the vector "lanes" (e.g., the 64-bit DP elements) where the corresponding bit in the predicate register is 'true' will execute the operation and update their destination register.
 - This is the primary mechanism for handling conditional logic (e.g., if statements) inside loops, which avoids the performance penalties of branch mispredictions.



- **Gather/Scatter**

- Hardware support for non-contiguous memory access.
- This is essential for HPC applications dealing with irregular data structures or arrays-of-structures (AoS).



- **AI/HPC Data Types**

- Hardware support for half-precision floating point (FP16) and 8-bit/16-bit integer (INT8/INT16) dot product instructions.

- **Four-Operand FMA**

- Standard Fused Multiply-Add (FMA) operation is destructive (e.g., $Z_0 \leftarrow Z_0 + Z_1 * Z_2$), overwriting one of the input registers.
- A non-destructive 4-operand FMA (e.g., $Z_0 \leftarrow Z_3 + Z_1 * Z_2$) is highly desirable for many algorithms.



- **Peak FLOPS**

- Two 512-bit FMA pipelines (FLA and FLB), each core can execute 2 FMAs/cycle. In double-precision (DP), a 512-bit vector holds 8 elements.
- This yields a throughput of: 2 pipelines x 8 DP elements/pipeline x 2 ops/FMA = 32 DP FLOPS/cycle
- At 2.2 GHz, this is 70.4 GFLOPS/core, 3.3792 TFLOPS for the 48 compute cores.

- **Peak L1D Bandwidth**

- The memory unit is configured to execute **two 512-bit (64-byte) SIMD loads per cycle or one 512-bit (64-byte) SIMD store per cycle**.
- At 2.2 GHz, this provides 256 GB/s of load bandwidth and 128 GB/s of store bandwidth.